



AI-Powered Data Analytics Internship

Week 4 Submission — Final Report

Consolidated Insights, Forecasted Demand, and Strategic Recommendations (Weeks 1–3)

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Executive Summary

This report is the Week 4 final deliverable for the Excelerate AI-Powered Data Analytics Internship, and it consolidates the three prior submissions — the Week 1 Data Understanding Report, the Week 2 Cohort Participation Analysis Report, and the Week 3 Trained Model & Prediction Report — into a single, decision-ready document for programme managers. It summarises historical performance and seasonal demand, presents the forecasting results and predictive model built in Week 3, ranks opportunity categories by both historical performance and model-predicted participation probability, and closes with a set of scalable, evidence-based strategic recommendations.

The dataset underlying this work covers 8,558 learner-opportunity records across 23 opportunities in five categories (Internship, Course, Event, Competition, Engagement), drawn from a genuinely international base of 71 countries and over 1,800 institutions. Across all three prior weeks of analysis, one finding stands out consistently: Internship opportunities dominate the platform by volume (63.3% of all applications) but convert to active participation at a far lower rate (21.5%) than every other category (84–97%). The Week 3 predictive model — a Random Forest classifier achieving 85.7% test accuracy and a ROC-AUC of 0.9225 — confirms this pattern statistically, showing that opportunity_category and the number of opportunities available within it together account for 71.3% of what actually drives participation, far outweighing any demographic factor.

Building on this, the report projects near-term demand, ranks all 23 live opportunities by predicted participation probability, and translates these findings into four concrete recommendations for Week 4 and beyond: restructuring the Internship pipeline into smaller, lower-friction modules; scheduling major opportunity launches around the platform's proven seasonal peaks; using the model's probability rankings to target promotional support where it is most needed; and building lightweight re-engagement flows for the learners the model currently under-predicts. Suggested methods for measuring the impact of each recommendation are included in Section 10.

Read together, these three weeks show a platform that is very good at attracting interest and comparatively less good at converting it once a learner has committed to a specific opportunity type. That distinction matters for Week 4 planning, because the fix is not a broader marketing push — the data suggests the platform already has more demand than it is currently converting — but a more deliberate redesign of how the highest-volume category is structured and staffed.

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1. Introduction and Objectives

Week 4 marks the final phase of this internship project. Where Weeks 1 through 3 each focused on a single stage of the analytics pipeline — understanding the data, exploring and engineering it, then training and validating a predictive model — Week 4 draws all three together into one strategic narrative aimed squarely at programme managers rather than at a technical audience.

The objectives set for this phase were to:

- Summarise historical performance, seasonal trends, and forecasted demand to establish strategic priorities and high-growth opportunity rankings.
- Analyse historical application patterns and dataset trends across the full Week 1–3 body of work.
- Evaluate quarterly seasonality and category-level demand behaviour.
- Project peak demand periods and directional future application volumes.
- Develop practical, scalable recommendations based on the Week 3 predictive model, accounting for a diverse learner base.
- Propose concrete ways to measure the impact of those recommendations once implemented.

The intended outcome, as set out in the Week 4 brief, is a comprehensive executive summary delivering clear demand forecasts, identified growth opportunities, and actionable strategic recommendations.

2. Methodology: Consolidating Weeks 1–3

Bringing three separate weekly reports together into one coherent narrative was done in three steps.

Step 1 — Gathering All Reports, Visualisations, and Model Outputs

All three prior deliverables were reviewed in full: the Week 1 Data Understanding Report (data dictionary, cleaning log, and historical/category summary statistics), the Week 2 Cohort Participation Analysis Report (feature engineering, exploratory analysis, quarterly trends, and forecasting signals), and the Week 3 Trained Model & Prediction Report (the Random Forest classifier, its evaluation against three alternative algorithms, and its opportunity-level probability rankings). Every chart and figure referenced in this report is either carried forward directly from one of those three submissions or rebuilt from the same underlying `processed_data.csv` file, so that this report stays fully traceable to its sources.

Step 2 — Reviewing Findings for Consistency and Clarity

One inconsistency surfaced during this review and is worth documenting rather than quietly resolving. The Week 1 report, based on an earlier cleaning pass, cites 8,543 cleaned records and 5,417 Internship applications (63.4%). Weeks 2 and 3 both use the final

processed_data.csv export of 8,558 records and 5,421 Internship applications (63.3%) — a difference of 15 records, most likely test/QA rows removed in a later cleaning pass. This report follows Weeks 2 and 3 and treats 8,558 records as the authoritative figure throughout, since it is the version the Week 3 predictive model was actually trained on. Beyond this single reconciled figure, the historical and category-level findings across all three reports are consistent with one another and with a direct re-check against processed_data.csv..

Step 3 — Highlighting the Strongest Insights

Three insights recur across all three reports and form the backbone of this final submission:

- Internship dominates by volume but underperforms on conversion — confirmed descriptively in Weeks 1 and 2, and statistically in Week 3, where opportunity_category alone explains 36.7% of the trained model's predictive weight.
- Q1 is consistently the platform's busiest period, and also its most competitive — sign-ups, applications, and rejections all peak together in Q1 (most visibly 2024 Q1), while Q2 combines the lowest volume with the highest participation rate of any quarter (96.1%).
- Programme structure outweighs learner demographics as a driver of participation — gender contributes just 0.64% of the Week 3 model's predictive weight, versus 71.3% for category and category-level opportunity count combined.

3. Historical Analysis Summary

This section summarises how learner sign-ups and applications have grown since the dataset began (late 2022) through the most recent complete quarter on record (Q1 2024), drawing on the Week 1 monthly analysis and the Week 2 quarterly rebuild.

Overall Growth Trajectory

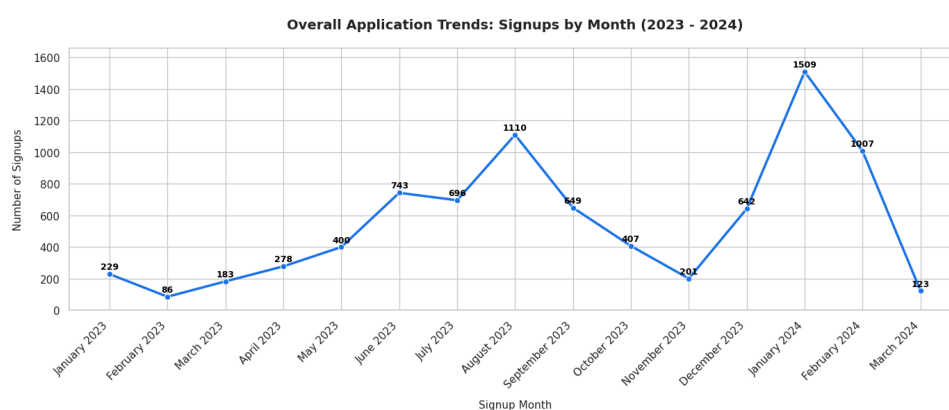


Figure 1. Overall monthly sign-up trend, 2023–2024 (Week 1).

Sign-ups grew steadily through 2023 — from 229 in January to a mid-year peak of 1,110 in August — eased through autumn, then surged to the highest point in the dataset, 1,509, in January 2024. March 2024's low count (123) reflects a partial month of data collection rather than a genuine drop in interest.

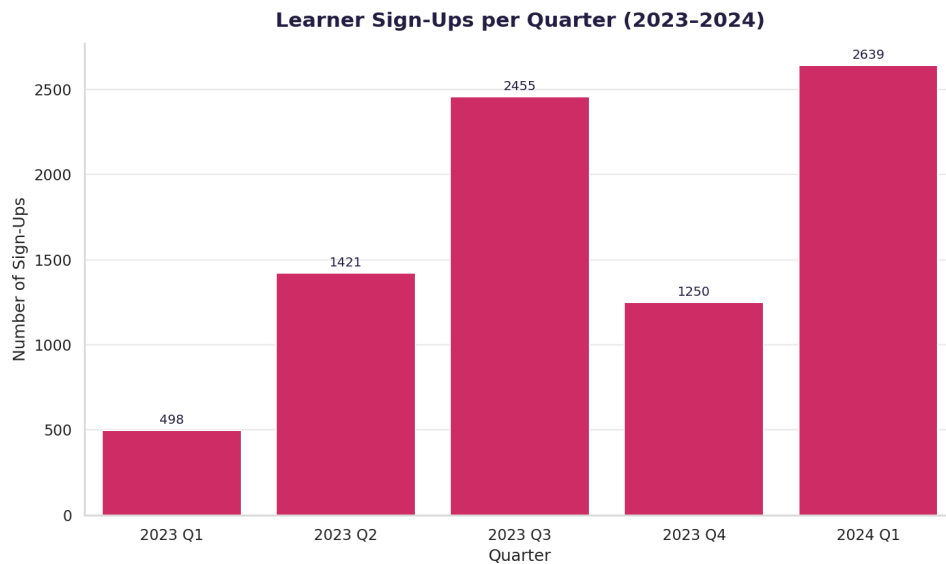


Figure 2. Sign-ups per quarter, rebuilt from processed_data.csv (Week 2).

Viewed by quarter rather than by month, the same growth story holds: sign-ups climbed from 498 in 2023 Q1 to 2,639 in 2024 Q1, with 2023 Q3 (2,455) the strongest quarter within 2023 itself.

Growth and Decline Patterns

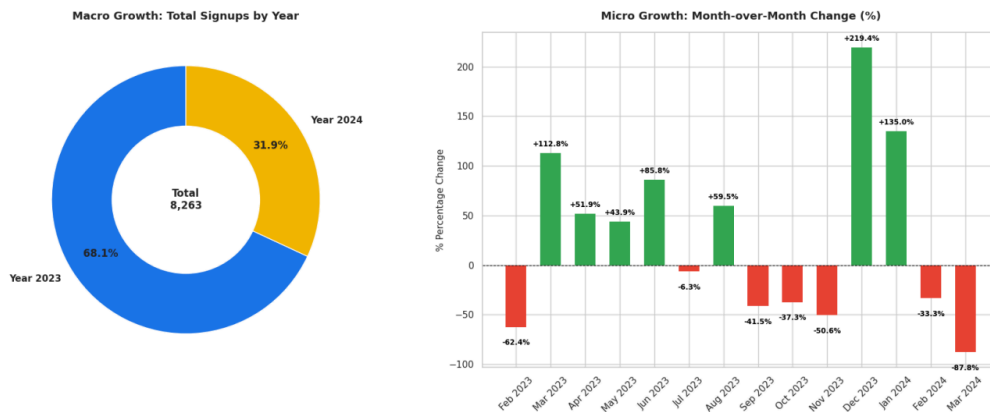


Figure 3. Yearly sign-up share and month-over-month percentage change (Week 1).

Growth did not move in a straight line — it swung sharply from month to month, with the steepest single-month increases in December (+219.4%) and January (+135.1%), and the steepest declines in February and March, each year. This repeating rhythm — strong at year-end and year-start, weak in the following February/March — is one of the clearest and most consistent patterns in the entire three-week analysis, and is the basis for the seasonal timing recommendation in Section 9.

Application Progression by Category (Historical)

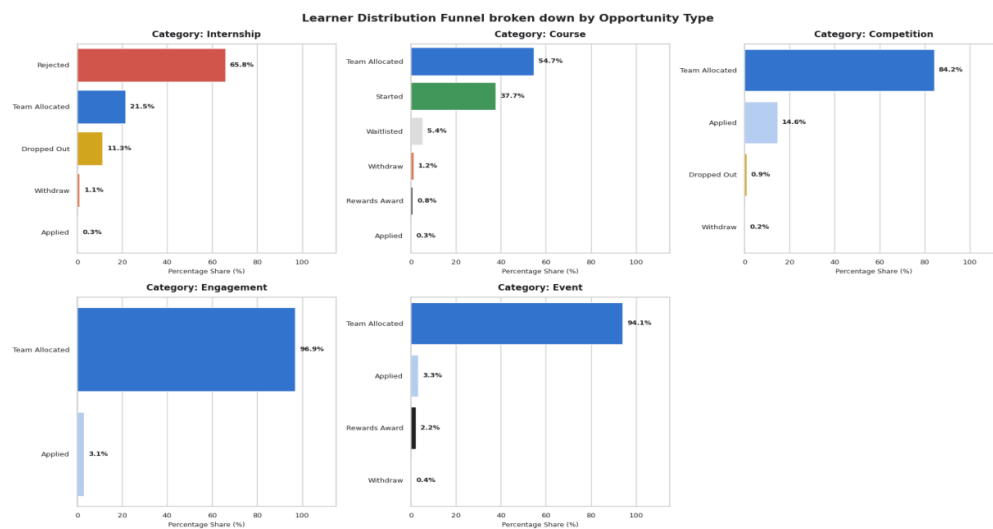


Figure 4. Status funnel breakdown for each opportunity category (Week 1).

This was the earliest appearance of the pattern that recurs throughout this report: Competition, Course, Engagement, and Event applicants overwhelmingly progress to an active status (84–97%), while 65.8% of Internship applicants were rejected outright, with only 21.5% reaching Team Allocated.

4. Seasonal Trend Analysis

This section evaluates quarterly seasonality and category-level demand behaviour in more depth, combining the Week 1 application-timing analysis with the corrected Week 2 quarterly breakdowns.

Quarterly Seasonality

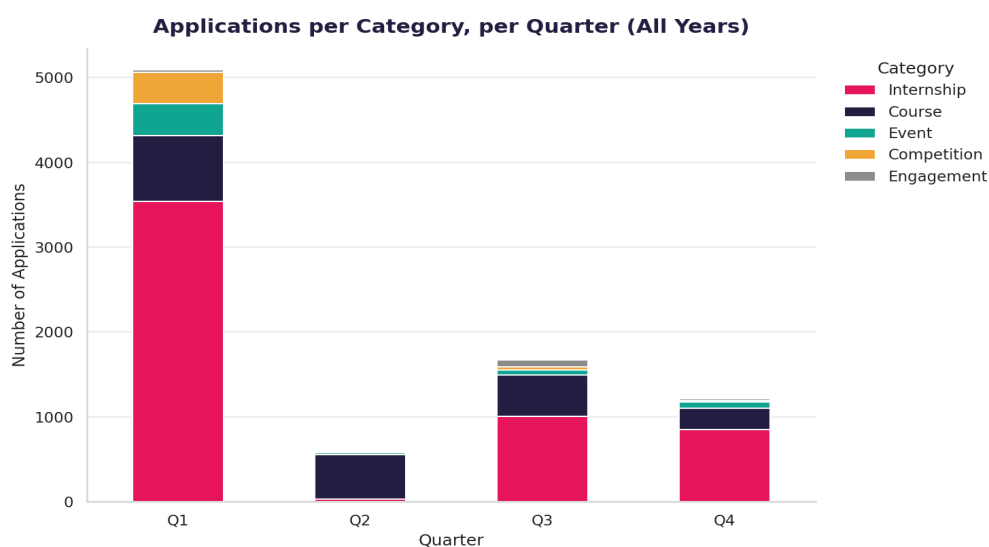


Figure 5. Applications per category, per quarter, all years combined (Week 2).

Q1 is the platform's busiest quarter for every major category, most dramatically for Internship. Q2 is consistently the quietest quarter by volume across the dataset.

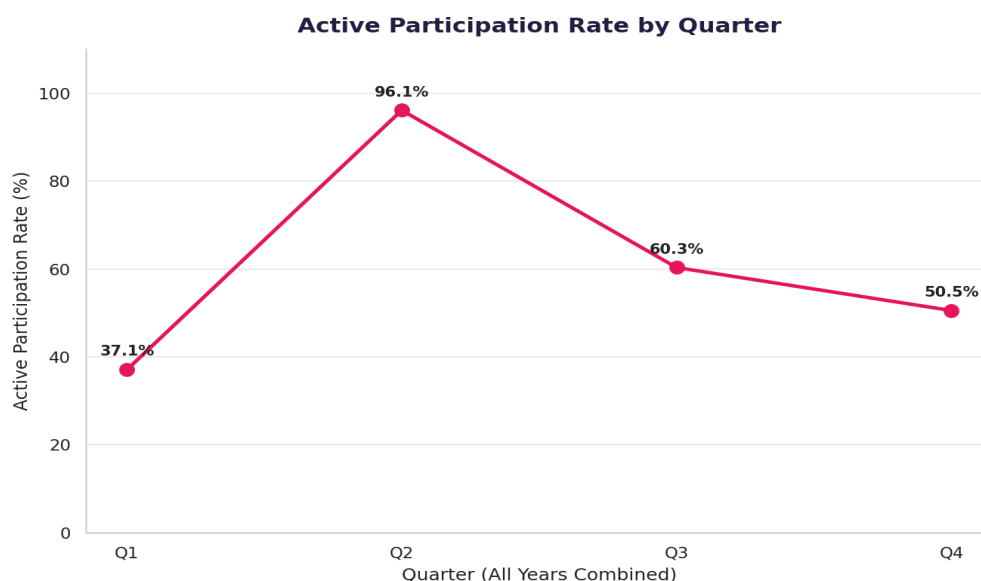


Figure 6. Active participation rate by quarter, all years combined (Week 2).

This is the other half of the seasonality story, and the more important half for planning purposes: the busiest quarter (Q1, 59% of all applications) has the lowest participation rate of any quarter (37.1%), while the quietest quarter (Q2, 589 applications) has the highest rate by a wide margin (96.1%). Volume and conversion move in opposite directions across the year.

Application Timing Behaviour

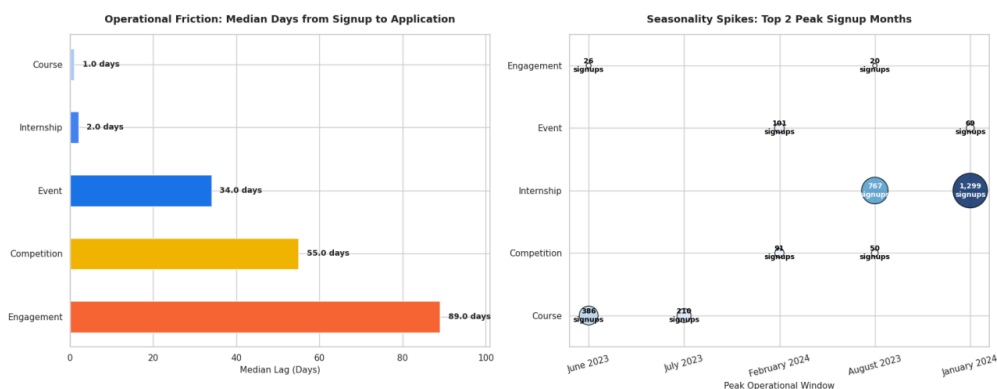


Figure 7. Median days from sign-up to application, and peak sign-up months, by category (Week 1).

Learners move almost immediately for Courses (median 1 day) and Internships (median 2 days), but take far longer to commit to Events (34 days), Competitions (55 days), and Engagement opportunities (89 days). Each category also peaks in a different month — Course and Engagement in June 2023, Internship in January 2024, and Competition/Event in February 2024 — which has a direct bearing on when each type of opportunity should be promoted.

Section Summary

- Q1 (particularly January) is the strongest acquisition and application window across the whole dataset, for sign-ups, applications, and — because Internship dominates this window — total rejections as well.
- Q2 is the platform's quiet quarter by volume but its strongest quarter by conversion (96.1% active participation), and deserves more attention as a proof-of-concept window for new or restructured opportunities.
- Q4 and the following Q1-to-Q2 transition (Feb–Mar) are consistently the weakest growth months, a pattern that held in both the Week 1 monthly view and the Week 2 quarterly rebuild.
- Different categories have very different "time to commit" profiles (1 day for Courses vs. 89 days for Engagement), so a single platform-wide promotional calendar is unlikely to be optimal — timing should be category-specific.

5. Opportunity Category Rankings

This section ranks the platform's five opportunity categories — and, within them, all 23 individual live opportunities — combining three independent sources of evidence: historical participation outcomes (Weeks 1–2), and the Week 3 Random Forest model's predicted participation probabilities. All three sources agree with one another, which gives programme managers a well-triangulated basis for prioritisation.

Historical Category Performance

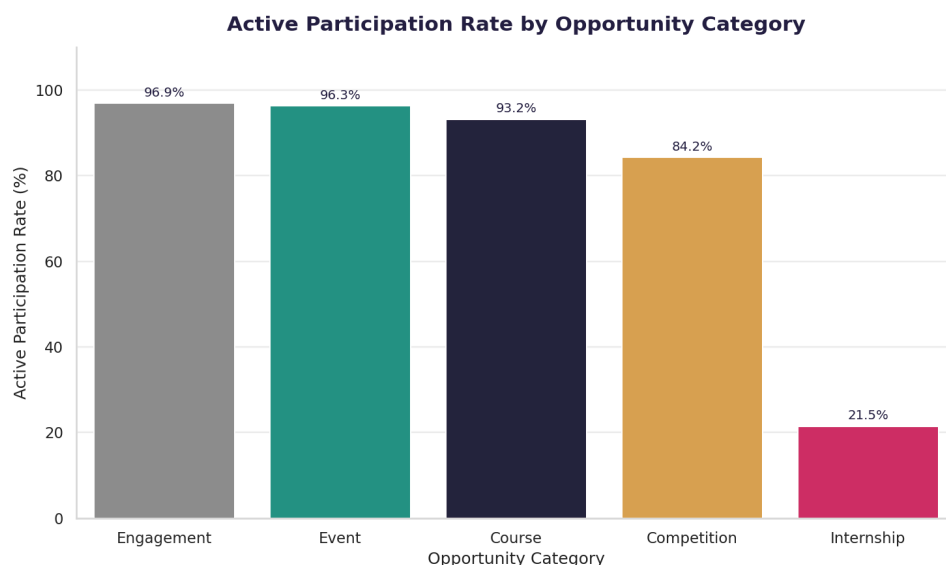


Figure 8. Active participation rate by opportunity category, historical (Week 2).

Category	Applications	% of Total	Active Participation Rate
Internship	5,421	63.3%	21.5%
Course	2,037	23.8%	93.2%
Event	545	6.4%	96.3%
Competition	425	5.0%	84.2%
Engagement	130	1.5%	96.9%

Because this table is built purely from historical outcomes, with no modelling involved, it serves as a useful sanity check against the predicted rankings that follow — if the two disagreed sharply, that would be a reason to distrust the model rather than the historical record.

Model-Predicted Category Rankings

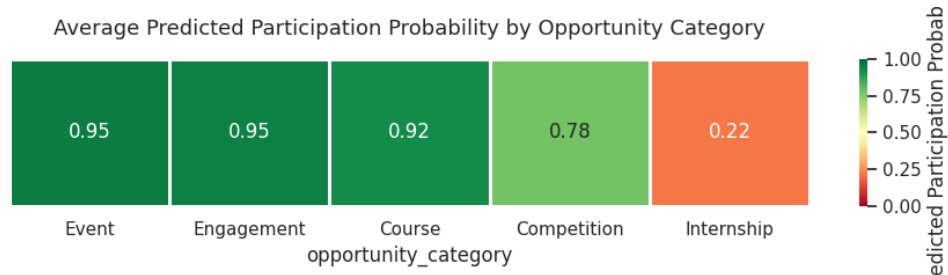


Figure 9. Model-predicted average participation probability, by category (Week 3).

The Random Forest model's predicted probabilities track the historical rates in the table above almost exactly (Event and Engagement both at 0.95, Course at 0.92, Competition at 0.78, Internship at 0.22) — a direct validation that the model has learned a genuine, well-calibrated pattern rather than an artefact of the training data.

That the model's predictions and the historical record converge this closely is, in a sense, the least exciting possible outcome for a data science exercise — and also the most reassuring one, since it means the model has not learned some idiosyncrasy of the training data that would fail to generalise to the next cohort.

Full Opportunity-Level Ranking

The Week 3 model scored all 23 live opportunities individually. Sorted from highest to lowest predicted participation probability, the ranking below is the most granular prioritisation tool available from the three weeks of work, and is the recommended starting point for Week 4 planning conversations.

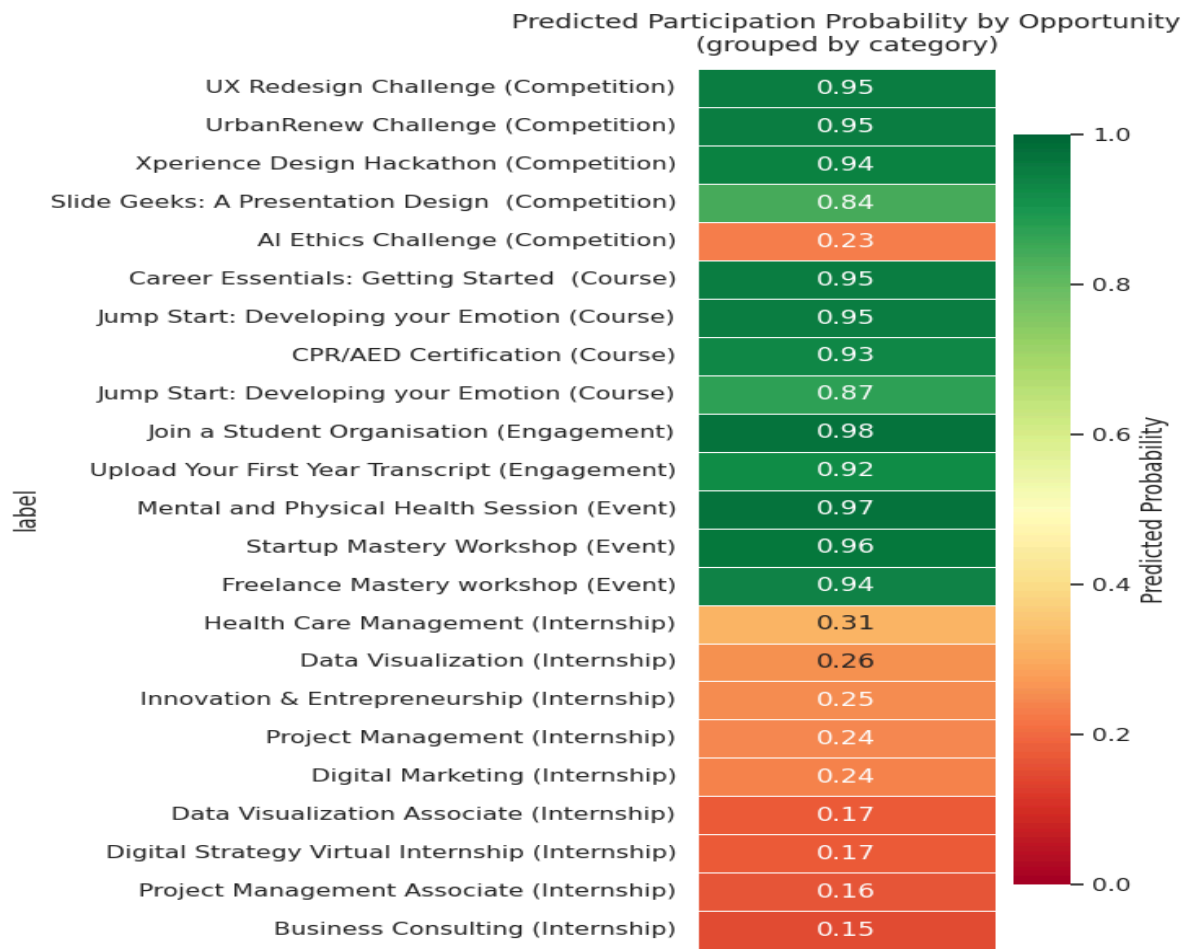


Figure 10. Predicted participation probability for every opportunity, grouped by category (Week 3).

#	Opportunity	Category	Pred. Prob.	Priority	Apps.
1	Join a Student Organisation	Engagement	97.5%	High	121
2	Mental and Physical Health Session	Event	96.6%	High	46
3	Startup Mastery Workshop	Event	96.2%	High	262
4	Career Essentials: Getting Started	Course	95.0%	High	1,423
5	Jump Start: Emotional Intelligence	Course	94.8%	High	33
6	UX Redesign Challenge	Competition	94.6%	High	112

#	Opportunity	Category	Pred. Prob.	Priority	Apps.
7	UrbanRenew Challenge	Competition	94.6%	High	138
8	Xperience Design Hackathon	Competition	94.2%	High	116
9	Freelance Mastery Workshop	Event	93.6%	High	237
10	CPR/AED Certification	Course	92.8%	High	389
11	Upload Your First Year Transcript	Engagement	92.2%	High	9
12	Jump Start: Emotional Intelligence (II)	Course	87.4%	High	192
13	Slide Geeks: Presentation Design	Competition	83.7%	High	4
14	Health Care Management	Internship	31.5%	Low	784
15	Data Visualization	Internship	25.6%	Low	980
16	Innovation & Entrepreneurship	Internship	25.0%	Low	419
17	Project Management	Internship	24.4%	Low	836
18	Digital Marketing	Internship	23.9%	Low	559
19	AI Ethics Challenge	Competition	22.8%	Low	55
20	Data Visualization Associate	Internship	17.3%	Low	551
21	Digital Strategy Virtual Internship	Internship	17.0%	Low	331
22	Project Management Associate	Internship	16.0%	Low	478

#	Opportunity	Category	Pred. Prob.	Priority	Apps.
23	Business Consulting	Internship	14.5%	Low	483

Thirteen of the 23 opportunities (56.5%) are High Priority, with predicted probabilities from 83.7% to 97.5%. The remaining ten — every one of them an Internship, apart from AI Ethics Challenge — are Low Priority, at 14.5% to 31.5%. The gap between these two tiers is stark and near-total: there is effectively no opportunity that sits in a genuine middle ground.

What This Ranking Confirms

- The category-volume paradox holds at the individual-opportunity level, not just in aggregate: all nine Internship listings individually score below 32%, while every non-Internship opportunity except one scores above 83%.
- High predicted probability does not require high application volume — Career Essentials (1,423 applications) and Slide Geeks (4 applications) both land in the High Priority tier, at 95.0% and 83.7% respectively.
- Within the Internship category, ranking still matters: Health Care Management (31.5%) clearly outperforms Business Consulting (14.5%), even though both sit in the Low Priority tier — useful for sequencing which Internship listings to redesign first.

6. Forecasting Results

This section brings together the Week 3 predictive model's performance and outputs with a directional volume outlook for the coming quarter, giving programme managers both a "who will participate" forecast and a "how much demand should we expect" forecast.

Predictive Model Summary

The Week 3 Random Forest classifier predicts whether a learner's application will result in active participation, using thirteen leakage-free features engineered from the Week 2 dataset. It was benchmarked against three alternative algorithms (Gradient Boosting, Decision Tree, Logistic Regression) on the same 6,846/1,712 train-test split before being selected as the final model.

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Random Forest (selected)	85.69%	90.48%	78.16%	83.87%	0.9225
Gradient Boosting	85.86%	90.64%	78.40%	84.08%	0.9221
Decision Tree	84.87%	87.37%	79.75%	83.39%	0.9138

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	82.71%	87.88%	73.87%	80.27%	0.8923

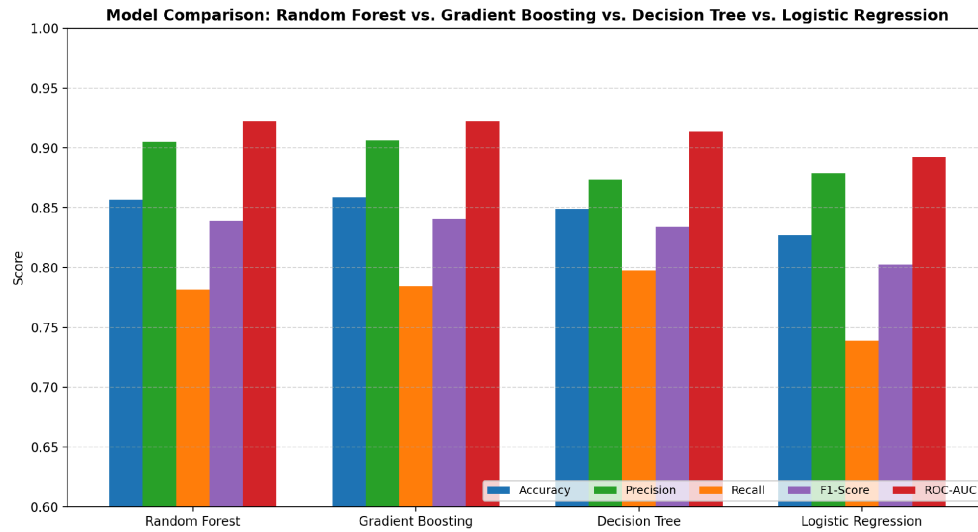


Figure 11. Comparison of all four evaluated models (Week 3).

Random Forest and Gradient Boosting form a clear top tier, essentially tied (ROC-AUC differs by just 0.0004). Random Forest was selected for its faster training time and lower overfitting risk, with Gradient Boosting retained as a strong alternative for future, larger datasets.

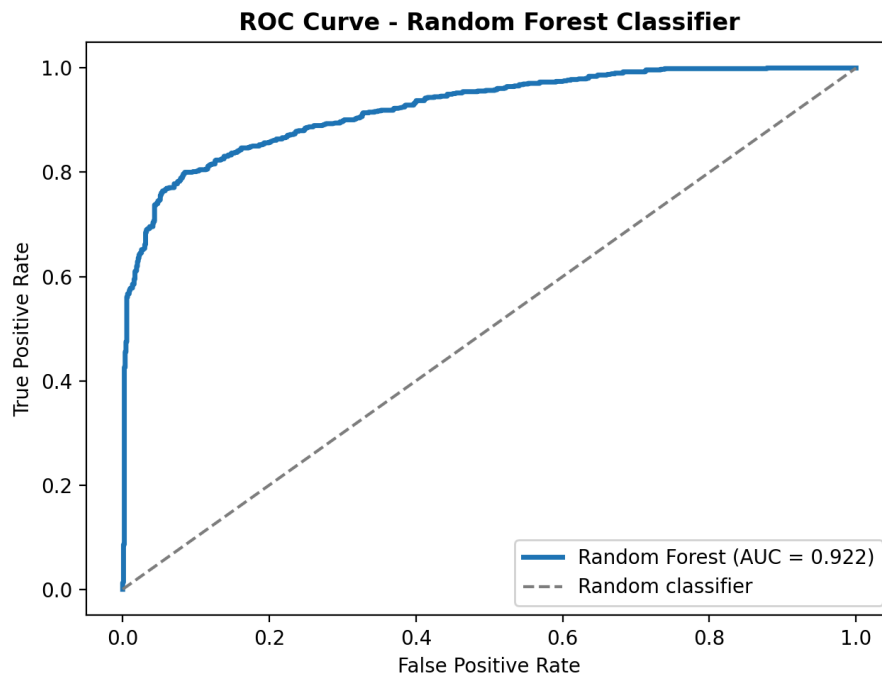


Figure 12. ROC curve for the selected Random Forest classifier (Week 3).

An AUC of 0.9225 means that, given one learner who went on to actively participate and one who did not, the model correctly ranks the active learner higher 92.25% of the time.

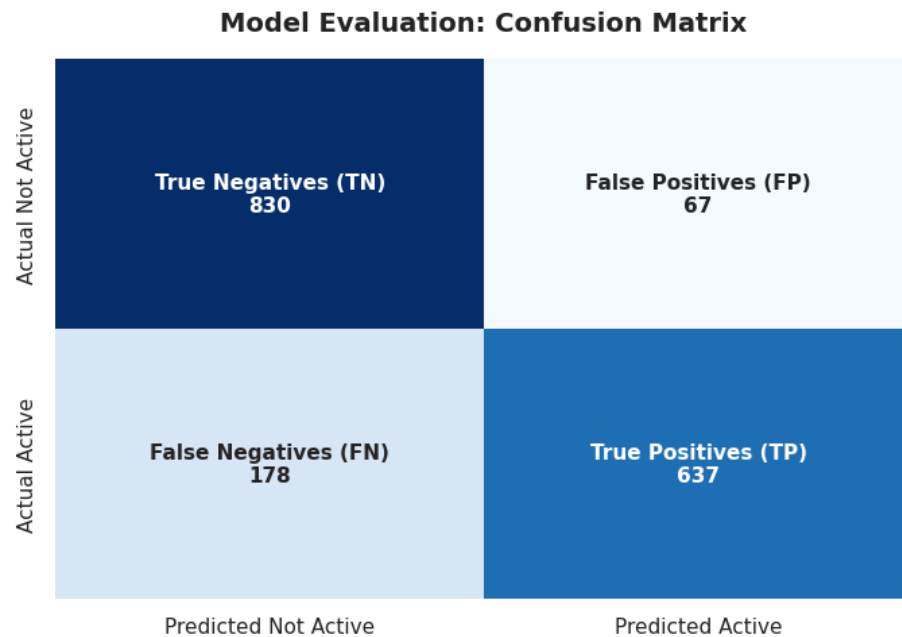


Figure 13. Confusion matrix on the 1,712-record held-out test set (Week 3).

The model is deliberately conservative: it rarely misclassifies an inactive learner as active (only 67 false positives, driving 90.5% precision), but it misses about one in five genuinely active learners (178 false negatives, 78.2% recall). Section 8 recommends a lightweight re-engagement flow specifically aimed at these under-predicted learners.

What Drives the Forecast: Feature Importance

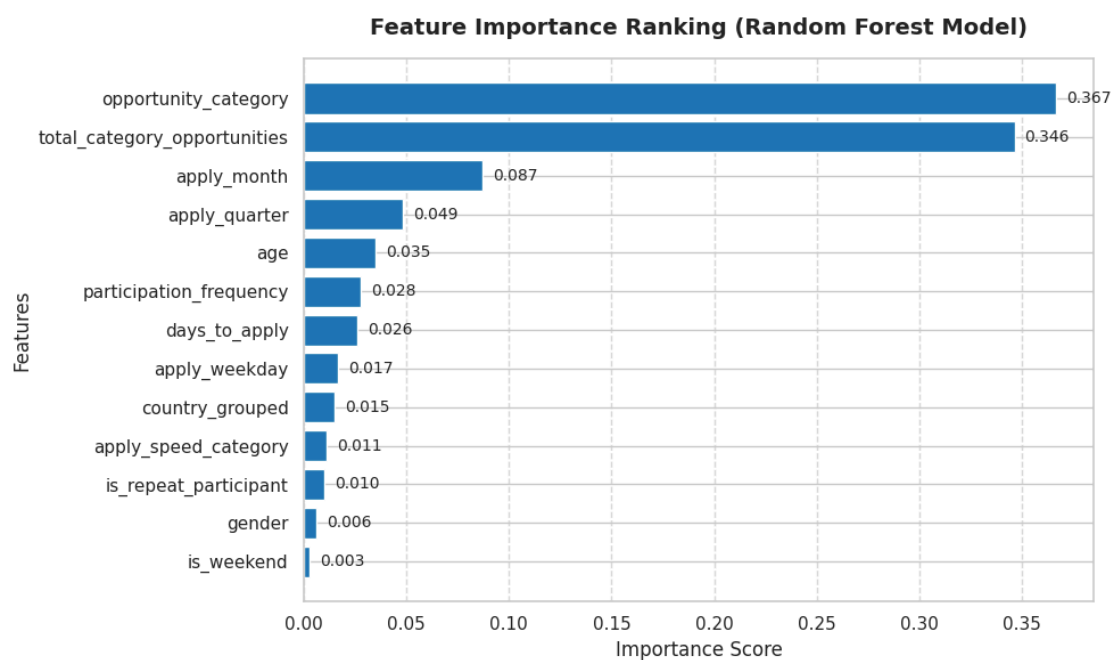


Figure 14. Feature importance ranking, Random Forest model (Week 3).

opportunity_category (36.7%) and total_category_opportunities (34.6%) together account for 71.3% of the model's predictive weight — by far the two most important drivers of whether a learner actively participates. Application timing (apply_month + apply_quarter) contributes a further 13.6%. Every demographic variable combined — gender, age, country — contributes less than 6%, with gender alone at just 0.64%. This is the clearest statistical confirmation in the whole three-week project that participation is a story about programme design and timing, not about who the learner is.

The practical consequence of this finding is that any future intervention aimed at improving participation should be judged first by whether it changes something about how an opportunity is categorised, structured, or timed. Interventions aimed at particular demographic groups are, on this evidence, unlikely to move the needle much on their own.

Demand Volume Outlook

Projecting exact future application volumes carries real uncertainty here: the dataset contains only one 2024 data point (Q1), and that quarter's volume (4,917 applications) was already more than the whole of 2023 combined (3,598), reflecting a genuine step-change in platform growth rather than a stable trend that can be safely extrapolated. Rather than presenting a single, falsely precise number, the chart below sets out a scenario range for 2024 Q2, built from the platform's own historical quarter-on-quarter swings (a –30% drop was recorded in 2023 Q4; a +30% rise is a conservative reading of the growth momentum seen elsewhere in the dataset).

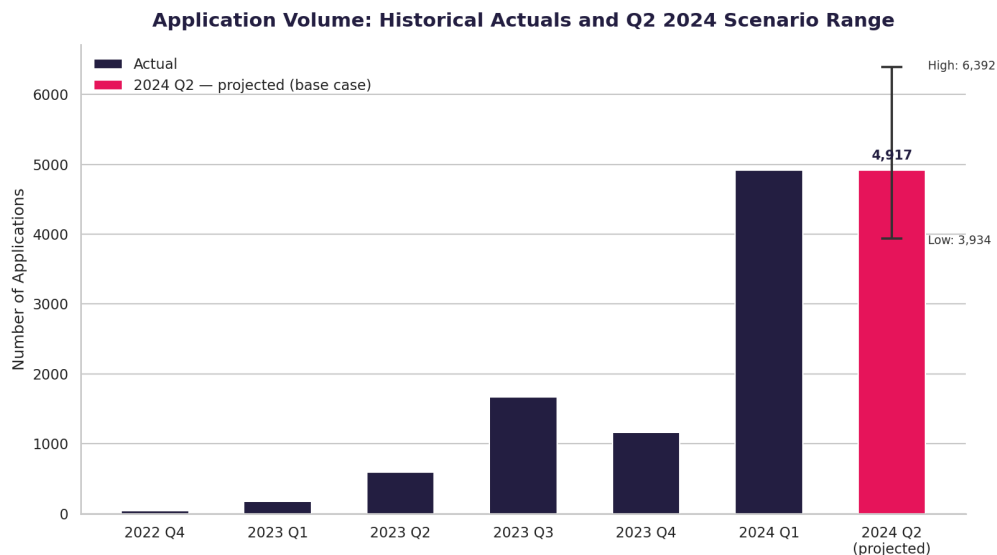


Figure 15. Historical quarterly application volumes and a scenario range for 2024 Q2.

Base case: roughly flat on Q1 2024 (~4,900 applications). Low case: a seasonal pull-back similar in scale to 2023 Q4 (~3,900). High case: continued growth momentum (~6,400). Given Q2 has historically been the platform's highest-converting quarter (96.1% active participation, per Section 4), even the low-volume scenario would be a strong outcome for programme managers if conversion quality holds.

Recommendation: rather than relying on this projection alone, the same Random Forest pipeline should be re-run once 2024 Q2 data is available, both to sharpen this forecast and to confirm the category-volume paradox continues to hold as the platform scales.

The honest caveat here is that a single quarter of 2024 data is a thin basis for any forecast, however carefully the range around it is constructed. This projection should be treated as a planning input to be revisited once real Q2 data lands, not a number to be held to precisely.

7. Dashboard & Visualisations

This section gathers the headline metrics from all three weeks into a single at-a-glance dashboard, followed by two supporting visuals — learner demographics and geography — that give programme managers the full context behind the numbers above.

Key Metrics at a Glance

Metric	Value	Source
Total learner-opportunity records	8,558	Week 2 / 3
Live opportunities	23, across 5 categories	Week 1 / 3
Countries represented	71	Week 1
Institutions represented	1,800+	Week 1
Overall active participation rate	47.6%	Week 2 / 3
Internship share of applications	63.3%	Week 2 / 3
Internship active participation rate	21.5%	Week 2 / 3
Non-Internship active participation rate (range)	84.2% – 96.9%	Week 2 / 3
Model test accuracy	85.69%	Week 3
Model ROC-AUC	0.9225	Week 3
Top 2 predictive features (combined weight)	opportunity_category + total_category_opportunities = 71.3%	Week 3
Busiest quarter (by volume)	2024 Q1 — 4,917 applications (57%)	Week 2
Highest-converting quarter	2023 Q2 — 96.1% active participation	Week 2

Learner Demographics

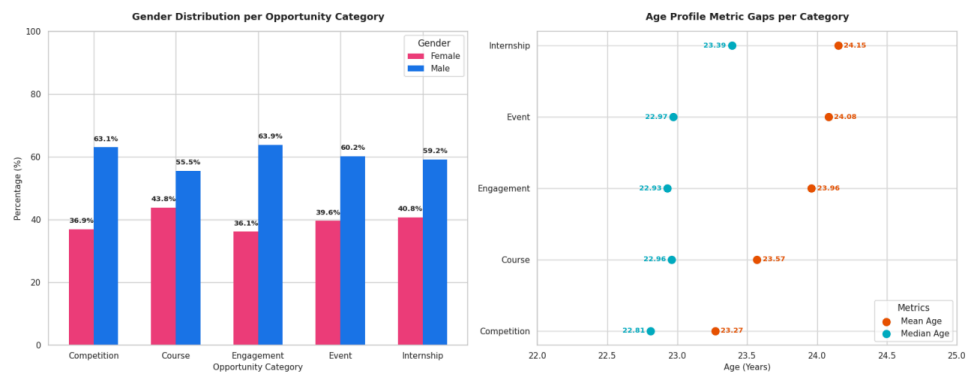


Figure 16. Gender distribution and age profile by opportunity category (Week 1).

Learners are consistently in their early twenties across every category (23–24 years on average) and skew male overall (roughly 59–64% depending on category), with Course showing the highest share of female learners (43.8%). As Section 6 shows, none of this meaningfully affects predicted participation — gender contributes only 0.64% of the model's predictive weight.

This is the same conclusion Section 6 reaches from a completely different angle — there, it is the model's own feature weights that show demographics barely matter; here, it is simply that the demographic profile does not vary enough across categories to explain the participation gap by itself.

Geographic and Institutional Reach

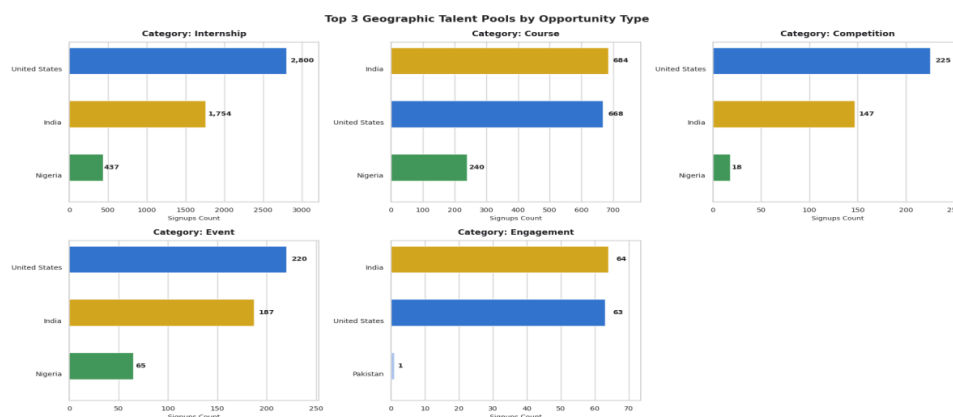


Figure 17. Top three contributing countries, by opportunity category (Week 1).

The United States and India are the two leading sources of learners in almost every category, with Nigeria a consistent third contributor — confirming the platform's genuinely international reach across all three weeks of analysis.

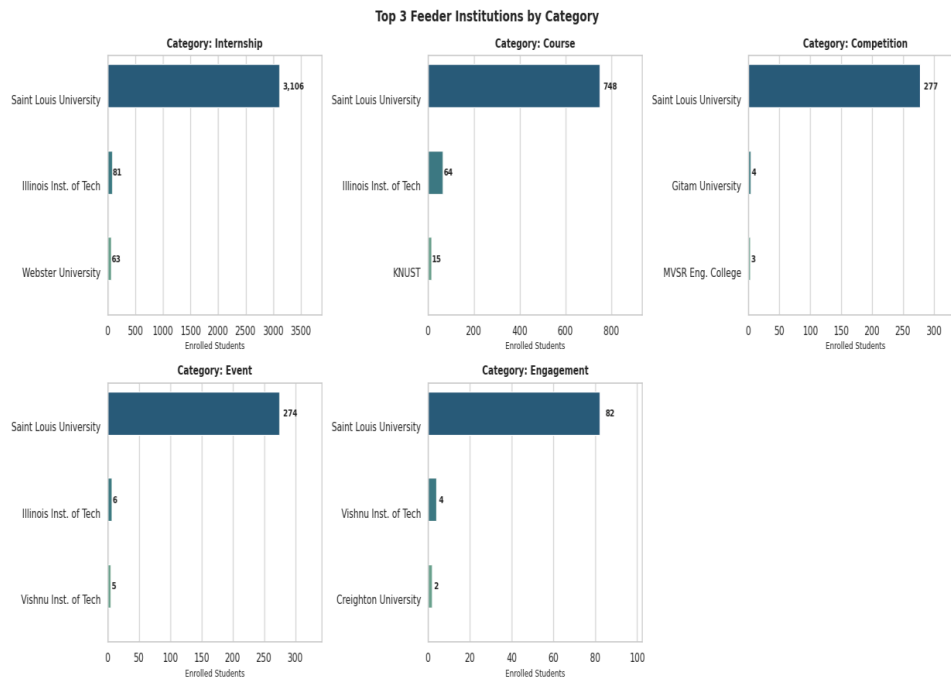


Figure 18. Top three feeder institutions, by opportunity category (Week 1).

Saint Louis University is the largest single source of learners in every category by a wide margin, with Illinois Institute of Technology the next-largest contributor for Course, Internship, and Event.

The concentration around a small number of institutions, Saint Louis University in particular, is worth flagging as a limitation as much as a finding: conclusions drawn from this dataset describe this specific learner population well, but should be applied cautiously if the platform expands into a meaningfully different institutional base.

8. AI-Assisted Analysis Documentation

AI tools supported every phase of this project, always under human direction and with every output independently validated against the underlying data before being included in a report. The table below consolidates how AI was used across all three weeks.

Week	Prompt / Request Used	Validation Approach
Week 1	Requests focused on generating and refining the Python code used for the initial data analysis, and on drafting and simplifying written interpretations of early findings for a non-technical reader.	Code was run and tested in a notebook environment; outputs were manually checked against the dataset; interpretations were reviewed for accuracy before being included in the report.
Week 2	“Analyze the provided dataset and the accompanying summary of findings from	Every AI-surfaced pattern was cross-referenced against

Week	Prompt / Request Used	Validation Approach
	the team’s exploratory data analysis. Focus on identifying non-obvious patterns, explaining the causes behind key trends, flagging any anomalies, and formulating practical business recommendations.”	the team’s own EDA and statistical summaries; key claims were manually recalculated from the raw data before being trusted.
Week 3	Requests centered on turning the team’s own model results and feature importances into a readable priority matrix, styling calibration and diagnostic charts for a non-technical audience, and drafting an executive synthesis of the findings into Week 4 recommendations.	Logic and calculations were independently reproduced; predicted probabilities were sanity-checked against historical baselines; overfitting was monitored via the train-test accuracy gap; code and column names were reviewed line by line.

Across all three weeks, the same validation discipline was applied consistently: AI-generated code was always executed and checked before being trusted, AI-generated claims were always checked against the raw dataset, and AI was used to accelerate analysis and communication rather than to replace the team's own statistical judgement.

The consistent thread across all three weeks' use of AI tools is that they accelerated the mechanical parts of the work — drafting code, generating a first pass at a chart, summarising a large table — while every substantive claim in this report was checked against the underlying dataset before being written down. That distinction is why this section exists at all: it is meant to make the process auditable, not to credit or discount a finding based on how it was produced.

9. Strategic Recommendations

The four recommendations below are drawn directly from the Week 3 model's findings, and each is written to be scalable across the platform's genuinely diverse learner base — spanning 71 countries, a range of institutions from large state universities to smaller colleges, and a mix of first-time and repeat applicants — rather than tailored to any single group. Each includes a concrete suggestion for measuring its impact, as requested for this phase.

Recommendation 1 — Modularise the Internship Pipeline

The finding: Internship is the platform's highest-volume category (63.3% of applications) but its lowest-converting one (21.5% active participation), while every other category converts at 84% or higher.

The action: Restructure Internship listings into smaller, milestone-driven modules or micro-internships, mirroring the shorter, lower-friction, higher-completion structure that

already works well for Course and Event opportunities (median time-to-apply of 1–2 days, 92–96% participation).

Why it scales: This is a structural change to how opportunities are packaged, not a change targeted at any particular learner group — it should benefit applicants regardless of country, institution, or background, consistent with the Week 3 finding that demographic variables carry almost no predictive weight (gender: 0.64%).

Measuring impact: Track the active-participation rate of newly modularised Internship listings against the current 21.5% category baseline over the next two quarters; a successful redesign should show a clear, sustained improvement without a corresponding rise in the rejection rate.

Because this is a structural change to the opportunity itself rather than a targeting change aimed at particular applicants, it is also the recommendation least dependent on getting audience segmentation right — which matters, given how little the model's own feature weights suggest demographic segmentation would help in the first place.

Recommendation 2 — Time-Phase Opportunity Launches to Seasonal Demand

The finding: Application timing (apply_month + apply_quarter) carries 13.6% of the model's predictive weight, and Q1 and Q2 show opposite volume/conversion profiles (Q1: highest volume, lowest conversion at 37.1%; Q2: lowest volume, highest conversion at 96.1%).

The action: Launch high-priority or newly redesigned opportunities in the seasonal windows already shown to convert well (Q2, and the December–January window identified in Section 3), and pair Q1 launches — when Internship demand will naturally spike — with additional application-support resourcing to handle the volume without a corresponding drop in learner experience.

Why it scales: Seasonal patterns were shown to hold consistently across categories and, in the Week 1 analysis, across countries — this is a platform-wide scheduling change rather than a segment-specific one.

Measuring impact: Compare the active-participation rate of opportunities launched inside vs. outside the recommended windows, controlling for category, over the next full seasonal cycle.

This recommendation is deliberately about when opportunities launch rather than when they are promoted afterward, since the timing patterns in Section 4 describe learner commitment behaviour that is already largely set by the time an application opens. The more effective lever is choosing the launch window itself.

Recommendation 3 — Target Support Using the Probability Rankings

The finding: The Week 3 model produces a specific, ranked predicted-probability score for all 23 live opportunities (Section 5), separating them cleanly into a High Priority tier

(83.7–97.5%) and a Low Priority tier (14.5–31.5%), with Competition showing the widest internal spread (22.8–94.6%) of any category.

The action: Use the ranked table in Section 5 to prioritise where promotional pushes, onboarding support, or application-readiness resources (such as the checklist proposed in the Week 3 report) are deployed — starting with the lowest-scoring Internship listings and the most variable Competition listings, rather than spreading engagement resources evenly across all 23 opportunities.

Why it scales: Because the ranking is opportunity-level rather than learner-level, it applies uniformly regardless of who is applying, and can be recalculated automatically as new opportunities are added.

Measuring impact: After deploying support to the bottom-ranked opportunities, re-score them with the same model and check whether their predicted (and eventually actual) participation probability has moved measurably closer to the High Priority tier.

Competition's unusually wide internal spread, from 22.8% to 94.6%, is a useful illustration of why category-level averages are not, on their own, a sufficient planning tool. Two opportunities in the same category can sit in entirely different priority tiers, which is exactly the granularity the opportunity-level ranking in Section 5 is meant to provide.

Recommendation 4 — Build a Re-Engagement Flow for Under-Predicted Learners

The finding: The model's confusion matrix (Section 6) shows 178 false negatives on the test set alone — genuinely active learners the model predicted would not participate, disproportionately learners with a thin prior participation history.

The action: Build a lightweight onboarding or nudge flow specifically for first-time or low-frequency applicants, since the model's conservatism means these are exactly the learners most likely to be missed by any automated prioritisation, however well the model performs on average.

Why it scales: This recommendation is explicitly a safeguard against a scalability failure mode — it exists to make sure that automating engagement decisions with the Week 3 model does not systematically disadvantage newer or less-active learners, whatever their background.

Measuring impact: Track the eventual actual participation rate of learners the model flagged as unlikely to participate but who received the re-engagement flow, against a comparable group who did not, to quantify how many of the model's false negatives can be recovered.

This recommendation is best understood as a deliberate hedge against the model's own known blind spot, rather than an attempt to improve its accuracy directly. Accepting that the model will always under-predict some genuinely active learners, and building a process to catch them, is a more realistic goal than trying to eliminate that blind spot through further tuning.

Recommendation Summary

#	Recommendation	Primary Metric to Track	Target Direction
1	Modularise the Internship pipeline	Internship active-participation rate	Up from 21.5% baseline
2	Time-phase launches to seasonal demand	Participation rate, in-window vs. out-of-window launches	In-window higher
3	Target support via probability rankings	Predicted probability of bottom-ranked opportunities	Movement toward High Priority tier
4	Re-engage under-predicted learners	Actual participation rate among flagged false negatives	Up vs. untreated comparison group

Each of these four recommendations is deliberately scoped to be testable within a single quarter, using data the platform already collects — none require new instrumentation or a long waiting period before the team can tell whether it is working.

10. Business Impact Assessment

This section frames the plausible business impact of the four recommendations above. Because none of these changes has been implemented yet, the figures below are framed as directional, evidence-based estimates rather than guarantees — each is explicitly tied back to a specific number already established elsewhere in this report.

Internship Modularisation (Recommendation 1)

Internship currently converts 5,421 applications into active participation at 21.5% (roughly 1,166 active learners). Every category with a lower-friction, modular structure converts at 84% or higher. Even a partial improvement — say, halving the gap between Internship's current rate and the platform's next-lowest category rate (Competition, 84.2%) — would move Internship's conversion rate to roughly 53%, more than doubling active Internship participation without any increase in application volume. This is, by a wide margin, the single highest-leverage change available to the programme, simply because of Internship's sheer scale.

It is worth being explicit that halving the conversion gap is an illustrative target rather than a guaranteed outcome — the real figure will depend on exactly how the modularisation is implemented — but even a more modest improvement, given Internship's scale, would still represent the largest single gain available anywhere in this dataset.

Seasonal Launch Timing (Recommendation 2)

2023 Q4's quarter-on-quarter decline (–49.1% in sign-ups) shows how much volume is plausibly being lost to poor timing alone. Shifting even a portion of major launches into the

Q2 and December–January windows already shown to convert well could lift overall conversion without requiring any new learner acquisition spend.

None of this requires acquiring a single new learner; it is entirely about when the platform makes its existing pipeline of opportunities available, which is one of the reasons this recommendation was prioritised alongside the higher-profile Internship redesign.

Targeted Prioritisation (Recommendation 3)

Concentrating limited engagement resources on the ten Low Priority opportunities (currently 14.5–31.5% predicted probability, representing 5,421 applications) rather than spreading support evenly across all 23 opportunities should produce a measurably larger return per unit of staff or marketing effort than the platform's current, undifferentiated approach.

The comparison worth drawing here is opportunity cost rather than raw numbers: every hour of engagement staff time spent on an opportunity already converting at 95% is an hour not spent on one converting at 15%, where the marginal return on that same hour is very likely larger.

Re-Engagement Flow (Recommendation 4)

Recovering even a modest share of the 178 test-set false negatives (a 10.4% false-negative rate on top of an already-conservative model) represents genuine participation the current process is at risk of missing — and, because this group is defined by low historical activity rather than by demographics, addressing it directly supports equitable access to the platform's opportunities.

This recommendation is smaller in scale than the other three, which is appropriate — it is a safeguard against a specific failure mode of automation, not a lever expected to move the platform's overall numbers on its own.

Overall Assessment

Taken together, these four recommendations point toward the same underlying opportunity: the platform's growth in learner acquisition (Section 3) has consistently outpaced its growth in conversion quality, most visibly in the Internship category. The single highest-impact action available for Week 4 and beyond is not attracting more applicants — sign-ups already grew more than five-fold across the period covered by this dataset — but converting more of the applicants the platform already has, by restructuring its highest-volume, lowest-converting category and by using the Week 3 model's outputs to allocate engagement effort where it will do the most good.

If this report has a single sentence worth remembering above all the others, it is this: the constraint on this platform's growth is not how many learners it can attract, but how well it converts the ones it already has — and that is a solvable, largely internal problem rather than one that depends on a bigger marketing budget.

11. Conclusion

This report has consolidated four months of work — from the first data-understanding pass in Week 1, through the exploratory and feature-engineering work of Week 2, to the trained and validated predictive model of Week 3 — into a single strategic narrative for programme managers. Every major finding across the three weeks points toward the same conclusion: active learner participation on this platform is driven overwhelmingly by how an opportunity is structured and when it is offered, not by who the learner is.

The category-volume paradox — Internship's combination of dominant volume and weak conversion — is the single most consequential and consistently-replicated finding in the entire project, confirmed descriptively in Week 1, quantified in Week 2, and statistically validated by a well-generalising machine learning model in Week 3 (85.7% accuracy, 0.9225 ROC-AUC, only a 1.43-point train-test gap). The four recommendations in Section 9, together with the measurement approach attached to each, give the team a concrete, evidence-based plan for turning that finding into improved outcomes over the coming quarters.